

BST Creation Algorithm to program

```
#include<stdio.h>
```

```
struct node
```

```
{  
    int info;  
    struct node* left;  
    struct node* right;
```

```
};
```

```
typedef struct node* nodeptr;
```

```
nodeptr getnode()
```

```
{  
    return (nodeptr)malloc(sizeof(struct node));  
}
```

```
nodeptr maketree(int x)
```

```
{  
    nodeptr p;  
    p=getnode();  
    p->info=x;  
    p->left=NULL;  
    p->right=NULL;  
    return (p);
```

```
}
```

```
void main()
```

```
{  
    nodeptr root,p,q;  
    int number;  
    char ch;  
    printf("Enter the root:");  
    scanf("%d",&number);  
    root = maketree(number);  
    while(scanf("%d",&number)!=EOF)  
    {  
        p=q=root;  
        while(number!=p->info && q!=NULL)  
        {  
            p=q;  
            if(number<=p->info)  
                q=p->left;  
            else  
                q=p->right;  
        }  
        if(number == p->info)  
            printf("\n%d is a duplicate value",number);
```

```

        else if(number<p->info)
            setleft(p,number);
        else
            setright(p,number);
    }
    printf("\nInorder:");
    inorder(root);
    printf("\nPreorder:");
    preorder(root);
    printf("\nPostorder:");
    postorder(root);
}
void setleft(nodeptr p,int x)
{
    if(p==NULL)
        printf("Cant insert");
    else if(p->left != NULL)
        printf("Invalid insertion");
    else
        p->left = maketree(x);
}
void setright(nodeptr p,int x)
{
    if(p==NULL)
        printf("Cant insert");
    else if(p->right != NULL)
        printf("Invalid insertion");
    else
        p->right = maketree(x);
}
void inorder(nodeptr p)
{
    if(p!=NULL)
    {
        inorder(p->left);
        printf("%d\t",p->info);
        inorder(p->right);
    }
}
void preorder(nodeptr p)
{
    if(p!=NULL)
    {

```

```
        printf("%d\t",p->info);
        inorder(p->left);
        inorder(p->right);
    }
}
void postorder(nodeptr p)
{
    if(p!=NULL)
    {
        inorder(p->left);
        inorder(p->right);
        printf("%d\t",p->info);
    }
}
```